Cosmological solutions with torsion effects in Polynomial Affine Gravity

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November 29, 2023

1 Introduction

Einstein's theory of General Relativity, it is a gravitational theory where the interactions is solely mediated by, the metric tensor [1–3]. Providing an ansatz compatible with the symmetries of the observable universe (cosmological principle), it is possible to model the Universe with Einstein's field equations [4].

With the introduction of the cosmological constant (see Ref. [5]) it is possible to obtain non-trivial dynamics to describe the evolution of the universe, leading to nine different scenarios, depending on the values of the geometrical factor κ (which characterize the geometry of the spatial foliation of the Universe), and the cosmological constant Λ. Although, the introduction of a cosmological constant does indeed introduce dynamics to Einstein's static universe, its time evolution does not provide a precise description of the current state of the universe. In order to have a more accurate description of the observable universe, it is required the introduction of the energy-momentum tensor, along with the state equation. By taking into account this new information into Einstein's Field Equations, it is possible to portray the observable universe. Moreover, it allow us to clearly distinguish three different epochs, *evolving* from a radiation era, to a matter era and finally to an accelerated expansion due to dark energy presence. This is the universe describe by Friedmann–Lemaitre–Robertson–Walker (FLRW), see Refs. [6–12], and provides the foundations for the standard model of cosmology ΛCDM.

Einstein's formulation its the best theory we have so far to describe gravitational interactions, and there is an agreement between the theoretical predictions with the observational data, to see a full review on its empirical evidence, see Ref. [13,14]. Perhaps, one of its more important theoretical predictions, it is the existence of gravitational waves, which were detected by by LIGO-Virgo collaborations [15,16]. Despite its success, there are still non-trivial problems related to its description of the geometry of space-time, specifically problems related to a quantization of gravity [17–20] and the necessity of the dark sector to explain the galaxy rotation curve speed [21,22], along with accelerated expansion [23].

These type of problems suggest, in principle, Einstein's theory of gravity must be an effective theory of another gravitational theory. In that sense, there

are extensions of general relativity such as where the Einstein-Hillbert action its replaced by an arbitrary function, named $f(\mathcal{R})$, which was first proposed in Ref. [24]. An interesting case of this type of gravity, its the Starobinsky's model [25] that allows to describe a universe without a singularity. Moreover, due to quantum corrections to general relativity, lead to a new contribution to the Einstein-Hillbert action due to curvature squared new term [26], which is a special case of $f(\mathcal{R})$ gravity.

Because the choice on function $f(\mathcal{R})$ it is an arbitrary one, there is an abundance of cosmological models, however, one could go even further and at a fundamental level, claim, that the metric tensor and the affine connection (a prior) must be independent fields, these are the type of modify models are known as *metric-affine* models, and its formalism it is known as *Palatini* formalism. There is an extensive collection on the literature related to metric-affine models [27–31], and an extensive/complete review can be found in Ref. [32].

In addition, its possible to built models of gravity with the affine connection as its fundamental field. The first affine model of gravity was proposed by Sir Eddington [33], and pure affine formalism were proposed in Refs. [34–36]. These type of affine formulations models of gravity, use the square-root of the Ricci tensor instead of the metric tensor to define the invariant volume element

There are other extensions to Einstein's theory of general relativity, for example Einstein-Cartan model [37,38], which uses the Einstein-Hillbert action to define gravitational interactions, with a non-symmetric connection by introducing non-trivial torsion effects. Moreover, the inclusion of extra dimensions where first proposed and studied by T. Kaluza and O. Klein in Refs. [39], specifically, uses a fifth dimension which allows to unify the gravitational with an electromagnetism forces. There are many gravitational models and within different classifications, an illustrative example of a so-called *map of modify gravity* can be found in Ref. [40].

In order to built a purely affine gravitational model of gravity, the metric tensor can not play any role in its formulation and in its building blocks, and this is where the Polynomial Affine Gravity enters. The Polynomial Affine Gravity its an alternative and purely affine model of gravity whose fundamental field is the affine connection instead of the metric tensor. The model has been studied in Refs. [41–47] and on this work, we explore the consequences of including torsion effects in its field equations in four-dimensions.

This paper is organized as follows: In Section 2, we present a brief overview on how to built the Polynomial Affine Gravity model, highlighting its features, and how is possible to still have two different emergent-metrics structures despite being a purely affine model. A complete scan of cosmological solutions to the field equations is presented in Section 3. In Section 4, we analyse and discuss the cosmological solutions. Final remarks are presented in Section 5.

2 Polynomial Affine Gravity

The Polynomial Affine Gravity is a model of gravity whose fundamental field is the affine connection $\hat{\Gamma}$,[1] whose irreducible fields can be decomposed into its symmetric and skew symmetric part

$$\begin{aligned}\hat{\Gamma}_\alpha{}^\beta{}_\gamma &= \hat{\Gamma}_{(\alpha}{}^\beta{}_{\gamma)} + \hat{\Gamma}_{[\alpha}{}^\beta{}_{\gamma]}, \\ &= \Gamma_\alpha{}^\beta{}_\gamma + \mathcal{B}_\alpha{}^\beta{}_\gamma + \delta^\beta_{[\gamma}\mathcal{A}_{\alpha]},\end{aligned} \tag{1}$$

the first term corresponds to the symmetric part of the connection $\hat{\Gamma}_{(\alpha}{}^\beta{}_{\gamma)} = \Gamma_\alpha{}^\beta{}_\gamma$, whereas the last two terms are related to the torsion tensor, the former is the traceless torsion $\mathcal{B}_\alpha{}^\beta{}_\gamma$ and the latter correspond to its vectorial part $\mathcal{A}_\alpha$.

Since the symmetric part of the connection does not preserve the invariance under diffeomorphisms, it enters into the action through the covariant derivative ∇^Γ. Additionally, the volume form is defined by the wedge product (which does not required the metric tensor to be well-defined) $\mathrm{d}V^{\alpha\beta\gamma\delta} = \mathrm{d}x^\alpha \wedge \mathrm{d}x^\beta \wedge \mathrm{d}x^\gamma \wedge \mathrm{d}x^\delta$.

Therefore, the fundamental ingredients in our affine model of gravity are ∇, $\mathcal{B}$, $\mathcal{A}$ and $\mathrm{d}V$. The action is built up using a sort of *dimensional analysis* technique. The method allows us to consider every single possible configuration of scalar densities composed by powers of all the irreducible fields of the affine connection [42, 45]. In the same spirit the three-dimensional model was also built up Ref. [46].

The most general action (up to topological invariants and boundary terms) in four dimensions is given by

$$\begin{aligned}S = \int \mathrm{d}V^{\alpha\beta\gamma\delta}\Big[&B_1\mathcal{R}_{\mu\nu}{}^\mu{}_\rho\mathcal{B}_\alpha{}^\nu{}_\beta\mathcal{B}_\gamma{}^\rho{}_\delta + B_2\mathcal{R}_{\alpha\beta}{}^\mu{}_\rho\mathcal{B}_\gamma{}^\nu{}_\delta\mathcal{B}_\mu{}^\rho{}_\nu + B_3\mathcal{R}_{\mu\nu}{}^\mu{}_\alpha\mathcal{B}_\beta{}^\nu{}_\gamma\mathcal{A}_\delta \\ &+ B_4\mathcal{R}_{\alpha\beta}{}^\sigma{}_\rho\mathcal{B}_\gamma{}^\rho{}_\delta\mathcal{A}_\sigma + B_5\mathcal{R}_{\alpha\beta}{}^\rho{}_\rho\mathcal{B}_\gamma{}^\sigma{}_\delta\mathcal{A}_\sigma + C_1\mathcal{R}_{\mu\alpha}{}^\mu{}_\nu\nabla_\beta\mathcal{B}_\gamma{}^\nu{}_\delta \\ &+ C_2\mathcal{R}_{\alpha\beta}{}^\rho{}_\rho\nabla_\sigma\mathcal{B}_\gamma{}^\sigma{}_\delta + D_1\mathcal{B}_\nu{}^\mu{}_\lambda\mathcal{B}_\mu{}^\nu{}_\alpha\nabla_\beta\mathcal{R}_\gamma{}^\lambda{}_\delta + D_2\mathcal{B}_\alpha{}^\mu{}_\beta\mathcal{B}_\mu{}^\lambda{}_\nu\nabla_\lambda\mathcal{B}_\gamma{}^\nu{}_\delta \\ &+ D_3\mathcal{B}_\alpha{}^\mu{}_\nu\mathcal{B}_\beta{}^\lambda{}_\gamma\nabla_\lambda\mathcal{B}_\mu{}^\nu{}_\delta + D_4\mathcal{B}_\alpha{}^\lambda{}_\beta\mathcal{B}_\gamma{}^\sigma{}_\delta\nabla_\lambda\mathcal{A}_\sigma + D_5\mathcal{B}_\alpha{}^\lambda{}_\beta\mathcal{A}_\sigma\nabla_\lambda\mathcal{B}_\gamma{}^\sigma{}_\delta \\ &+ D_6\mathcal{B}_\alpha{}^\lambda{}_\beta\mathcal{A}_\gamma\nabla_\lambda A_\delta + D_7\mathcal{B}_\alpha{}^\lambda{}_\beta\mathcal{A}_\lambda\nabla_\gamma A_\delta + E_1\nabla_\rho\mathcal{B}_\alpha{}^\rho{}_\beta\nabla_\sigma\mathcal{B}_\gamma{}^\sigma{}_\delta \\ &+ E_2\nabla_\rho\mathcal{B}_\alpha{}^\rho{}_\beta\nabla_\gamma\mathcal{A}_\delta + F_1\mathcal{B}_\alpha{}^\mu{}_\beta\mathcal{B}_\gamma{}^\sigma{}_\delta\mathcal{B}_\mu{}^\lambda{}_\rho\mathcal{B}_\sigma{}^\rho{}_\lambda + F_2\mathcal{B}_\alpha{}^\mu{}_\beta\mathcal{B}_\gamma{}^\nu{}_\lambda\mathcal{B}_\delta{}^\lambda{}_\rho\mathcal{B}_\mu{}^\rho{}_\nu \\ &+ F_3\mathcal{B}_\nu{}^\mu{}_\lambda\mathcal{B}_\mu{}^\nu{}_\alpha\mathcal{B}_\beta{}^\lambda{}_\gamma\mathcal{A}_\delta + F_4\mathcal{B}_\alpha{}^\mu{}_\beta\mathcal{B}_\gamma{}^\nu{}_\delta\mathcal{A}_\mu\mathcal{A}_\nu\Big].\end{aligned} \tag{2}$$

In the above action, the covariant derivative and the curvature tensor are defined with respect to the symmetric part of the connection, meaning $\nabla = \nabla^\Gamma$ and $\mathcal{R} = \mathcal{R}^\Gamma$.

One of the most important features of this model its the lack of the metric tensor in its formulation causes that the number of terms in the action is finite, we called this property *rigidity*.

[1] The action is polynomial in the connection and its derivative

Additionally, even though, the manifold is endowed with an affine structure $\hat{\Gamma}$, it is still possible to have emergent metric structures. The first one comes from the Ricci tensor $\mathcal{R}_{\mu\nu}$, which is defined by the contraction of the Riemann tensor.[2]

$$\hat{\Gamma} \to \nabla^{\Gamma} \to \mathcal{R}_{\alpha\beta}{}^{\gamma}{}_{\delta} \to \mathcal{R}_{\beta\delta}. \tag{3}$$

A second metric structure, comes from the contraction of the product of two torsion tensor.[3] This idea was first introduce by Poplwaski in Ref. [48], and the metric structure its defined as follow

$$\mathcal{P}_{\alpha\delta} = \left(\mathcal{B}_{\alpha}{}^{\beta}{}_{\gamma} + \delta^{\beta}_{[\gamma}\mathcal{A}_{\alpha]}\right)\left(\mathcal{B}_{\beta}{}^{\gamma}{}_{\delta} + \delta^{\gamma}_{[\delta}\mathcal{A}_{\beta]}\right). \tag{4}$$

Interestingly, all the coupling constant are dimensionless, which suggest that the model is power-counting renormalizable. This, is a necessary condition but no sufficient condition to guarantee that the Polynomial Affine Gravity is renormalizable. Moreover, it suggest a conformal symmetry.

Moreover, in the hypothetical scenario of its quantization, because of its *rigidity*, all possible counter-terms will have the form of the ones already written in eq.(2). This is something desirable from a stand point of Quantum Field Theory view, for the reason that the superficial degree of divergence vanishes.

Finally, in the torsion-free sector it is possible to recover all Einstein's vacuum solutions. Even more, it is possible to coupled a scalar field using the *dimensional analysis* technique with the torsion tensor and the volume form, while the potential energy re-scale the volume element. From which, there is a subspace of solutions that allow us to recover Einsteins's field equations coupled with a scalar field, see Ref.[Paper˙Inflacion]

Since we are interested in the study of the cosmology, we want to impose the symmetries of the cosmological principle in our fundamental field $\hat{\Gamma}$ and in its irreducible fields. In order to build up the ansatz we compute the Lie derivative of each irreducible field along the Killing vectors ξ_i that generates the symmetries of homogeneity (translations) $\mathcal{P}_i$ and isotropy (rotations) $\mathcal{J}_i$. An explicit computation of the Lie derivative can be found in Ref. [49]. In what follows we shall show only the results.

The computation of the Lie derivative of Γ along the Killing vectors, determined its coefficients:

$$\Gamma_t{}^t{}_t = f(t), \qquad \Gamma_i{}^t{}_j = g(t)S_{ij}, \tag{5}$$

$$\Gamma_t{}^i{}_j = h(t), \delta^i_j \qquad \Gamma_i{}^j{}_k = \gamma_i{}^j{}_k, \tag{6}$$

where S_{ij} is the three-dimensional rank two symmetric tensor defined as follow

$$S_{ij} = \begin{pmatrix} \frac{1}{1-\kappa r^2} & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{pmatrix}, \tag{7}$$

[2]The Riemann tensor its defined by the commutator of covariant derivatives acting on a vector, therefore, it does not required the existence of a metric structure.

[3]This comes from the antisymmetric part of the affine connection

and γ is the symmetric connection compatible with desired symmetries written as

$$\gamma_r{}^r{}_r = \frac{\kappa r}{1-\kappa r^2}, \qquad \gamma_\theta{}^r{}_\theta = \kappa r^3 - r, \quad \gamma_\varphi{}^r{}_\varphi = \left(\kappa r^3 - r\right)\sin^2\theta, \quad \gamma_r{}^\theta{}_\theta = \frac{1}{r}, \tag{8}$$

$$\gamma_\varphi{}^\theta{}_\varphi = -\cos\theta\sin\theta, \quad \gamma_r{}^\varphi{}_\varphi = \frac{1}{r}, \qquad \gamma_\theta{}^\varphi{}_\varphi = \frac{\cos\theta}{\sin\theta}. \tag{9}$$

Additionally, the affine function $f(t)$ can be set equal to zero, under a parametrization of the time coordinate, for more information on this type of transformation, refer to Ref. [46]. Therefore, there are only two non trivial functions to define completely the symmetric part of the connection.

Following a similar procedure for the torsion tensor, is possible to determined the ansatz compatible with the required symmetries. For its traceless part $\mathcal{B}$, the non-vanishing components are

$$\mathcal{B}_\theta{}^r{}_\varphi = \psi(t) r^2 \sin\theta\sqrt{1-\kappa r^2}, \quad \mathcal{B}_r{}^\theta{}_\varphi = \frac{\psi(t)\sin\theta}{\sqrt{1-\kappa r^2}}, \quad \mathcal{B}_r{}^\varphi{}_\theta = \frac{\psi(t)}{\sqrt{1-\kappa r^2}\sin\theta}, \tag{10}$$

while the non-trivial components of its vectorial part $\mathcal{A}$ is

$$\mathcal{A}_t = \eta(t). \tag{11}$$

Using the cosmological ansatz presented in eq. (5), the Ricci tensor is defined as follow

$$\mathcal{R}_{tt} = \dot{h} + h^2 \qquad \mathcal{R}_{rr} = \frac{\dot{g} + gh + 2\kappa}{1-\kappa r^2}, \tag{12}$$

moreover, the Poplawski's metric is computed using eqs. (10) and (11)

$$\mathcal{P}_{tt} = \eta^2 \qquad \mathcal{P}_{rr} = -\frac{2\psi^2}{1-\kappa r^2}. \tag{13}$$

In order to use the Ricci tensor $\mathcal{R}_{\mu\nu}$ or the Poplawski $\mathcal{P}_{\mu\nu}$ as an emergent metric tensor, is necessary that the tensor is well defined, meaning, it can be invertible. If the tensor are well behaved, then the factors $(\dot{g} + gh + 2\kappa)$ or $\left(-2\psi^2\right)$ play an analogue role to the scale factor $a^2(t)$.

Finally, the field equations are obtained by varying the action with respect to each irreducible field using Kijowkski's formalism, see Ref. [50], which we also reviewed in Ref. [45]. The complete set of the field equation for each irreducible field can be found in Ref. [51].

3 Cosmological solutions

Using the ansatz found in eqs. (5), (10) and (11) the field equation are

$$\left(B_3\left(\dot{g} + gh + 2\kappa\right) - 2B_4\left(\dot{g} - gh\right) + 2D_6\eta g - 2F_3\psi^2\right)\psi = 0, \tag{14}$$

$$\left(B_3\eta\psi - 2B_4\eta\psi + C_1\left(\dot{\psi} - 2h\psi\right)\right)g = 0, \tag{15}$$

$$\left(B_3 + 2B_4\right)\eta g\psi + 2C_1\left(\kappa\psi + 4gh\psi - g\dot{\psi} - \psi\dot{g}\right) + 2\psi^3\left(2D_2 - D_1 - D_3\right) = 0, \tag{16}$$

$$\begin{aligned} &B_3\left(\eta\left(h\psi - \dot{\psi}\right) - \psi\dot{\eta}\right) - 2B_4\left(\eta\left(-h\psi - \dot{\psi}\right) - \psi\dot{\eta}\right) \\ &+ C_1\left(4h^2\psi + 2\psi\dot{h} - \ddot{\psi}\right) + D_6\eta^2\psi = 0, \end{aligned} \tag{17}$$

$$\begin{aligned} &B_3\left(\dot{g} + gh + 2\kappa\right)\eta - 2B_4\left(\dot{g} - gh\right)\eta + C_1\left(2\kappa h + 4gh^2 + 2g\dot{h} - \ddot{g}\right) \\ &+ 6h\psi^2\left(2D_2 - D_1 - D_3\right) + D_6\eta^2 g - 6F_3\eta\psi^2 = 0 \end{aligned} \tag{18}$$

where $\mathcal{A}$ and $\mathcal{B}$ have only one equation, eq. (14) and eq. (18) respectively, whereas there are three field equations for Γ, which are eqs. (15) to (17). Notice, we have four unknown functions of time $g(t)$, $h(t)$, $\psi(t)$ and $\eta(t)$ while there are five differential equations, meaning the system is overdetermined.

Nonetheless, we shall show that, it can actually be solved analytically without any assumption. To see this, we develop a *logical scheme* that allow us to seek (systematically) for solutions. First, notice eqs. (14) and (15) are of the form

$$\mathcal{F}(g, \dot{g}, h, \psi, \eta)\psi = 0, \tag{19}$$

$$\mathcal{G}(h, \psi, \dot{\psi}, \eta)g = 0, \tag{20}$$

where

$$\mathcal{F}(g, \dot{g}, h, \psi, \eta) = B_3\left(\dot{g} + gh + 2\kappa\right) - 2B_4\left(\dot{g} - gh\right) + 2D_6\eta g - 2F_3\psi^2, \tag{21}$$

$$\mathcal{G}(h, \psi, \dot{\psi}, \eta) = B_3\eta\psi - 2B_4\eta\psi + C_1\left(\dot{\psi} - 2h\psi\right). \tag{22}$$

Thus, from eqs. (19) and (20) its possible to distinguish four different branches:

$$\mathcal{F}(g, h, \psi, \eta) = 0 \quad \wedge \quad \mathcal{G}(h, \psi, \eta) = 0, \tag{23}$$

$$\mathcal{F}(g, h, \psi, \eta) = 0 \quad \wedge \quad g = 0, \tag{24}$$

$$\mathcal{G}(h, \psi, \eta) = 0 \quad \wedge \quad \psi = 0, \tag{25}$$

$$\psi = 0 \quad \wedge \quad g = 0. \tag{26}$$

In terms of the richness of the solutions, the most interesting branch its eq. (23), which is the most general case, and the simplest branch will be eq.(26).

3.1 First branch

This branch has the most general case, where the field equations are

$$B_3\left(\dot{g}+gh+2\kappa\right)-2B_4\left(\dot{g}-gh\right)+2D_6\eta g-2F_3\psi^2=0, \tag{27}$$

$$B_3\eta\psi-2B_4\eta\psi+C_1\left(\dot{\psi}-2h\psi\right)=0, \tag{28}$$

$$\left(B_3+2B_4\right)\eta g\psi+2C_1\left(\kappa\psi+4gh\psi-g\dot{\psi}-\psi\dot{g}\right)+2\psi^3\left(2D_2-D_1-D_3\right)=0, \tag{29}$$

$$\begin{aligned}&B_3\left(\eta\left(h\psi-\dot{\psi}\right)-\psi\dot{\eta}\right)-2B_4\left(\eta\left(-h\psi-\dot{\psi}\right)-\psi\dot{\eta}\right)\\&+C_1\left(4h^2\psi+2\psi\dot{h}-\ddot{\psi}\right)+D_6\eta^2\psi=0,\end{aligned} \tag{30}$$

$$\begin{aligned}&B_3\left(\dot{g}+gh+2\kappa\right)\eta-2B_4\left(\dot{g}-gh\right)\eta+C_1\left(2\kappa h+4gh^2+2g\dot{h}-\ddot{g}\right)\\&+6h\psi^2\left(2D_2-D_1-D_3\right)+D_6\eta^2g-6F_3\eta\psi^2=0\end{aligned} \tag{31}$$

From this, we solve eq.(28) to find an expression for $\eta(t)$

$$\eta(t)=\left(\frac{2h\psi-\dot{\psi}}{\psi}\right)\left(\frac{C_1}{B_3-2B_4}\right), \tag{32}$$

replacing the above expression for $\eta(t)$ into eq. (30), leads to two sub-branches for the $h(t)$ function

$$h(t)=\frac{\dot{\psi}}{2\psi} \qquad h(t)=\frac{\dot{\psi}}{\psi}\left(\frac{C_1D_6}{3B_3^2-8B_3B_4+B_4^2+2C_1D_6}\right) \tag{33}$$

using the simplest form of $h(t)$ function, then eq. (29) leads to

$$-2\left(D_1-2D_2+D_3\right)\psi^3+2C_1\left(\psi\left(\kappa-\dot{g}\right)+g\dot{\psi}\right)=0, \tag{34}$$

which is a first order differential equation which can be solve for $g(t)$ in terms of the ψ function

$$g(t)=\psi(t)\left(g_0+\int_1^t\left(\frac{\kappa}{\psi(\tau)}-\psi(\tau)\left(\frac{D_1-2D_2+D_3}{C_1}\right)\right)\mathrm{d}\tau\right) \tag{35}$$

the above solution also solves eq. (31), where g_0 is an integration constant. Then, eq. (27) becomes an integro-differential equation of first order

$$\begin{aligned}&\dot{\psi}\left(g_0+\int_1^t\left(\frac{\kappa}{\psi(\tau)}-\psi(\tau)\left(\frac{D_1-2D_2+D_3}{C_1}\right)\right)\mathrm{d}\tau\right)\left(\frac{3B_3-2B_4}{2}\right)\\&-\psi^2\frac{\left(B_3-2B_4\right)\left(D_1-2D_2+D_3\right)+2C_1F_3}{C_1}+\kappa\left(3B_3-2B_4\right)=0.\end{aligned} \tag{36}$$

The above equation can be solved for the special case $\kappa = 0$, with the variable change $\psi(t) = \dot{\phi}(t)$, then

$$\ddot{\phi}\left(g_0 - \phi\alpha\right)\beta - \dot{\phi}^2\gamma = 0, \tag{37}$$

where

$$\alpha = \left(\frac{D_1 - 2D_2 + D_3}{C_1}\right) \qquad \beta = \left(\frac{3B_3 - 2B_4}{2}\right) \tag{38}$$

$$\gamma = \frac{\left(B_3 - 2B_4\right)\left(D_1 - 2D_2 + D_3\right) + 2C_1F_3}{C_1}, \tag{39}$$

whose solution

$$\phi(t) = \frac{g_0}{\alpha} + \frac{\left(\phi_0\left(\alpha\beta + \gamma\right)\left(t + \phi_1\right)\right)^{\frac{\alpha\beta}{\alpha\beta+\gamma}}}{\alpha\beta}, \tag{40}$$

where ϕ_0 and ϕ_1 are integration constant. From this

$$\psi(t) = \phi_0\left(\phi_0\left(\alpha\beta + \gamma\right)\left(t + \phi_1\right)\right)^{-\frac{\gamma}{\alpha\beta+\gamma}}. \tag{41}$$

From the above expression and using the relations in eqs. (32), (33) and (35) it is straightforward to find the rest of the affine functions

$$\eta(t) = 0, \tag{42}$$

$$h(t) = -\frac{\gamma}{2\left(\alpha\beta + \gamma\right)\left(t + \phi_1\right)}, \tag{43}$$

$$g(t) = \phi_0\left(\left(\alpha\beta + \gamma\right)\phi_0\left(t + \phi_1\right)\right)^{-\frac{\gamma}{\alpha\beta+\gamma}}\left(g_0 - \frac{\left(\left(\alpha\beta + \gamma\right)\phi_0\left(t + \phi_1\right)\right)^{\frac{\alpha\beta}{\alpha\beta+\gamma}}}{\beta}\right) \tag{44}$$

3.2 Second branch

The second branch, imposes the $g(t) = 0$ restriction, which leads to a reduced field equations

$$2\kappa B_3\psi - 2F_3\psi^3 = 0, \tag{45}$$

$$g = 0, \tag{46}$$

$$2\kappa C_1\psi - 2\psi^3\left(D_1 - 2D_2 + D_3\right) = 0 \tag{47}$$

$$\kappa\left(4h^2C_1 + h\eta\left(B_3 + 2B_4\right) + D_6\eta^2 + 2C_1\dot{h} - \dot{\eta}\left(B_3 - 2B_4\right)\right) = 0, \tag{48}$$

$$2h\left(\kappa C_1 - 3\left(D_1 - 2D_2 + D_3\right)\psi^2\right) + 2\eta\left(\kappa B_3 - 3F_3\psi^2\right) = 0. \tag{49}$$

From eq. (45) its possible to find an expression for $\psi(t)$

$$\psi(t) = \pm\sqrt{\frac{\kappa B_3}{F_3}}, \tag{50}$$

then, replacing the above algebraic expression in eq.(49) leads to a relation between $h(t)$ and $\eta(t)$ as follow

$$h(t) = -\frac{2B_3F_3\eta}{3B_3\left(D_1 - 2D_2 + D_3\right) - C_1F_3}, \tag{51}$$

because of the above relation between the functions, eqs. (47) becomes a constraint equation for the coupling constant

$$\kappa\left(B_3\left(D_1 - 2D_2 + D_3\right) - C_1F_3\right) = 0 \rightarrow D_1 = \frac{2B_3D_2 - B_3D_3 + C_1F_3}{B_3}, \tag{52}$$

the above restriction, turns eq. (48) into a first order differential equation for $\eta(t)$

$$\dot{\eta}C_1\kappa\left(2B_4 - 3B_3\right) + \eta^2\left(3B_3^2 - 2B_3B_4 + C_1D_6\right)\kappa = 0 \tag{53}$$

which can be solved by

$$\eta(t) = \frac{C_1\left(2B_4 - 3B_3\right)}{\left(3B_3 - 2B_4\right)\left(tB_3 + \eta_0C_1\right) + tC_1D_6}, \tag{54}$$

then, the other the $h(t)$ can be written as follow

$$h(t) = \frac{B_3\left(3B_3 - 2B_4\right)}{\left(3B_3 - 2B_4\right)\left(tB_3 + \eta_0C_1\right) + tC_1D_6} \tag{55}$$

It is worth to notice the above non-trivial solution it is valid only for the case $\kappa \neq 0$ this can bee seen directly from eqs. (45) (52) and (53).

If one, impose the constraint $\kappa = 0$, then from eq. (45) tell us that $\psi(t) = 0$ which solves entirely the other equations. Therefore, the other functions remain unknown.

3.3 Third branch

With the constraint $\psi(t) = 0$, impose a strong restriction on the systems of differential equations, reducing its complexity to just one differential equation

$$g\eta^2D_6 + 2B_4\eta\left(gh - \dot{g}\right) + B_3\eta\left(2\kappa + gh + \dot{g}\right) + C_1\left(2h\left(\kappa + 2gh\right) + 2g\dot{h} - \ddot{g}\right) = 0 \tag{56}$$

the above differential equation has three unknown functions of time $h(t)$, $g(t)$ and $\eta(t)$ which can not be solved. The system for this branch undetermined.

3.4 Fourth branch

The fourth branch, impose strong restriction on the field equations by the requirements $g(t) = 0$ and $\psi(t) = 0$, the field equations are reduced to just one algebraic equation

$$\kappa \left(hC_1 + B_3\eta\right) = 0 \tag{57}$$

the above equation can be solved by setting $\kappa = 0$ then the functions $h(t)$ and $\eta(t)$ are undetermined, or by the relation $h(t) = -\eta(t)\left(\frac{B_3}{C_1}\right)$ where $\eta(t)$ its an arbitrary function.

4 Analysis of the solutions

The only non-trivial solutions can be found in branch (23), thus, this is the only branch which will be studied in the following subsection. In order to simplify our calculations, we are going to assume the values of the integration constant $\phi_0 \to 1$, $\phi_1 \to 0$ and $g_0 \to 0$. Thus, the affine functions are simplify up to

$$h(t) = \frac{\gamma}{2t\left(\alpha\beta + \gamma\right)} \qquad g(t) = -\alpha^2\beta\left(t\left(\alpha\beta + \gamma\right)\right)^{\frac{2\alpha\beta}{\alpha\beta+\gamma}-1}, \tag{58}$$

$$\psi(t) = \alpha\beta\left(t\left(\alpha\beta + \gamma\right)\right)^{-\frac{\gamma}{\alpha\beta+\gamma}} \qquad \eta(t) = 0, \tag{59}$$

where $h(t)$ and $g(t)$ defines the symmetric part of the connection, whereas $\psi(t)$ and $\eta(t)$ defines the anti symmetric part.

4.1 Emergent metrics

As mentioned in Sec. 1 it is possible to obtain emergent metric structures in this purely affine model of gravity. The first metric tensor, comes from the Ricci tensor, using eqs. (58) in eqs. (12) we compute the Ricci tensor

$$\mathcal{R}_{tt} = -\frac{3\gamma\left(2\alpha\beta + 3\gamma\right)}{4t^2\left(\alpha\beta + \gamma\right)^2}, \tag{60}$$

$$\mathcal{R}_{rr} = \Sigma\left(\alpha, \beta, \gamma\right) t^{2\left(\frac{-\gamma}{\alpha\beta+\gamma}\right)}, \tag{61}$$

where $\Sigma\left(\alpha, \beta, \gamma\right)$ is defined as

$$\Sigma\left(\alpha, \beta, \gamma\right) = -\frac{\alpha^2\beta\left(2\alpha\beta - 3\gamma\right)\left(\alpha\beta + \gamma\right)^{\frac{2\alpha\beta}{\alpha\beta+\gamma}}}{2\left(\alpha\beta + \gamma\right)^2}. \tag{62}$$

It is worth to recall that the Ricci's tensor spatial component plays an analogue role to the scale factor in the FRW universe, thus

$$\Sigma\left(\alpha, \beta, \gamma\right) t^{2\left(\frac{-\gamma}{\alpha\beta+\gamma}\right)} \longleftrightarrow a^2(t). \tag{63}$$

Therefore, we can use eq.(61) to define the notion of distance, as long as its well behaved, however, unlike in general relativity, the metriciy condition its

not satisfied in the affine geometry, meaning $\nabla\mathcal{R}$ does not vanishes, and as a consequence of this, there are non-metricity effects $\mathcal{Q}^{\mathcal{R}}_{\mu\nu\lambda}$ coming from the non-trivial effects related to the torsion tensor.

Additionally, it is also possible to compute Poplwaski's metric-torsion tensor which was defined in eq.(13) using eqs.(59), leading to

$$\mathcal{P}_{tt} = 0 \qquad\qquad \mathcal{P}_{rr} = \alpha^2\beta^2\left(t\left(\alpha\beta+\gamma\right)\right)^{-\frac{2\gamma}{\alpha\beta+\gamma}}, \tag{64}$$

however, unlike the Ricci tensor, the Poplwaski metric-torsion tensor has a trivial component in its temporal part, and, as a result of this, its a degenerate tensor which can not be invertible, therefore, it can not act as a well behaved metric tensor, and it is not possible to used to the define the notion of distance. Moreover, just like the Ricci tensor, its covariant derivative $\nabla\mathcal{P}$ does not vanishes, and, therefore there are non-metricity effects coming from the torsion tensor $\mathcal{Q}^{\mathcal{P}}_{\mu\nu\lambda}$.

4.2 Matter emulation from torsion

Having a well defined affine scale factor coming from the Ricci tensor as in eq. (61), we can see that this is a solution *sort-off* like a power-law solution, which depends on the coupling constant α, β and γ. In order to have a physical meaning, we can demand certain values of this parameters to have well known solutions.

Notice that $\psi(t)$ has a similar form to the energy density expression in the standard model of cosmology

$$\psi(t) = \alpha\beta\left(t\left(\alpha\beta+\gamma\right)\right)^{-\frac{\gamma}{\alpha\beta+\gamma}} \longleftrightarrow \rho(t) = \rho_0 a^{-3(1+\omega)} \tag{65}$$

, meaning we could use the torsion tensor to have different types of affine scales factor. For example, fixing the $\gamma = -\frac{\alpha\beta}{3}$, turns the affine scale factor

$$a_f(t) = a_0 t^{1/2}, \tag{66}$$

where a_0 its defined as

$$a_0 = \alpha^2\hat{\beta} \tag{67}$$

with $\hat{\beta} = -\beta^3$, where β must be a negative number in order to have a real affine scale factor, which translate into an inequality for the coupling constants

$$3B_3 < 2B_4. \tag{68}$$

This type of solution allow us to emulate the radiation era. The affine radiation era can be plotted for different parameter's values which satisfy the constraint condition, see Fig. 1.

In the same spirit, we can fix $\gamma = -\frac{2\alpha\beta}{5}$ which leads to

$$a_f(t) = a_0 t^{2/3}, \tag{69}$$

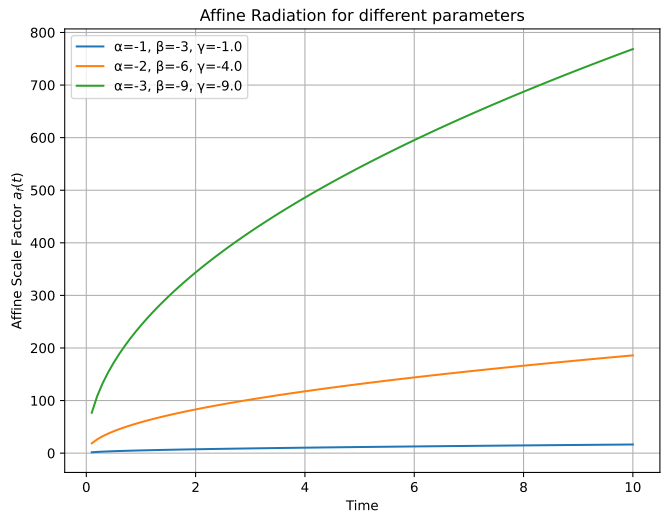

Figure 1: Radiation era in PAG due to torsion effects

where we use a_0 as

$$a_0 = \sqrt{-\alpha^{13/3}\beta^{10/3}}\left(\frac{2^{3/2}3^{2/3}}{5^{7/6}}\right) \tag{70}$$

which, in order to be real we need to impose the restriction $\beta < 0$, meaning, we use the same constraint written in eq.(68). This type of solution allow us to emulate a non-relativistic matter era. A plot for different values of the parameters can be found in fig. 2.

If we want to have an accelerated expansion due to torsion effects, we need to impose

$$\ddot{a}_f > 0 \rightarrow \frac{\gamma\left(\alpha\beta + 2\gamma\right)}{\left(\alpha\beta+\gamma\right)^2} t^{\frac{\alpha\beta}{\alpha\beta+\gamma}-3} > 0. \tag{71}$$

the above condition, ensures a positive acceleration. In order to have a steady growth acceleration, it is required that

$$\frac{\alpha\beta}{\alpha\beta+\gamma} - 3 > 0, \tag{72}$$

which leads to

$$\alpha\beta < -\frac{3}{2}\gamma, \tag{73}$$

this conditions ensures that the time evolution will be defined positive. Additionally, we have two branches coming from the numerator and denominator of eq. (71)

$$B1 : \alpha\beta > -2\gamma \qquad\qquad \gamma > 0, \tag{74}$$

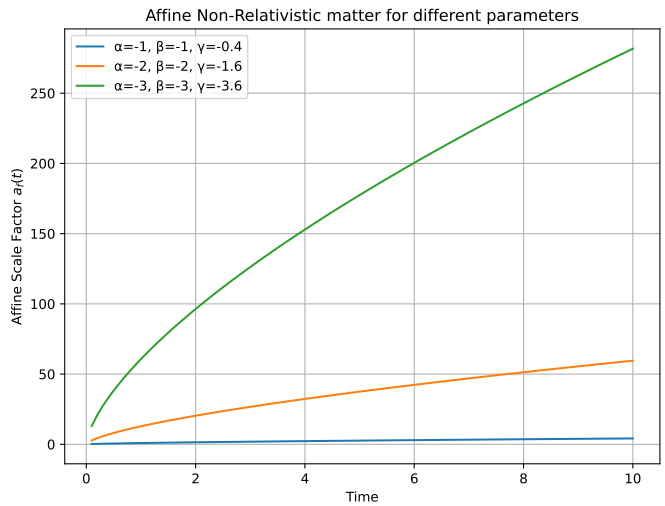

Figure 2: Non-relativistic matter era in PAG due to torsion effects

$$B2 : \alpha\beta < -2\gamma \qquad\qquad \gamma < 0. \tag{75}$$

Using eq. (73) along with eqs.(74) (75) allow us to define to intervals for the constant $\alpha\beta$, the former are $\alpha\beta \in \left[-2\gamma, -\frac{3}{2}\gamma\right]$, whereas the latter is given by $\alpha\beta < -2\gamma$. A plot for different values of the parameters can be found in fig. 3.

5 Final remarks

In this work, we have reviewed the Polynomial Affine Gravity model with torsion effects. The inclusion of the non-vanishing torsion tensor, allows for an overdetermined system of differential equations, see eqs. (14) to (18), whereas without torsion effects, the field equations are reduced to the *harmonic curvature* $\nabla_\alpha \mathcal{R}_{\beta\gamma}{}^{\alpha}{}_{\delta} = 0$, which through Bianchi's identity can be written as an anti symmetrized covariant derivative of the Ricci tensor $\nabla_\mu \mathcal{R}_{\alpha\beta} = 0$.

Without torsion effects, the field equation leads to one differential equation for two unknown functions, see eq.(18) with $\eta(t) = 0$, thus, the system is undetermined and, in order to extract information, its necessary to provide with an ansatz for $h(t)$ or $g(t)$. A priori, there is no physical reason to fix one of the functions, however, one could explore subspace of this torsion-free solutions, by studying $\nabla_\alpha \mathcal{R}_{\beta\gamma} = 0$. This, leads to a three differential equations, which can be solved analytically and provides with an exact solution for $h(t)$ and $g(t)$ functions. Even more, the Ricci tensor its well behaved and it can be used as a metric tensor, moreover, this type of solution its known as de-Sitter/Anti

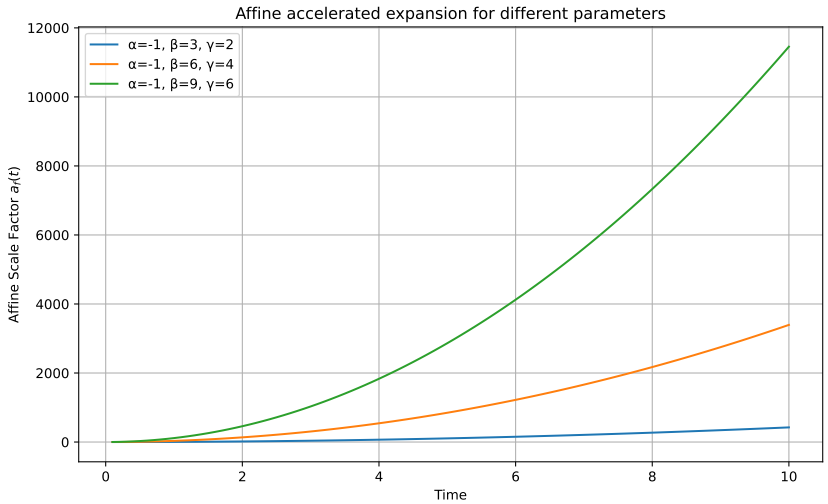

Figure 3: Accelerated expansion in PAG due to torsion effects

de-Sitter solution, where the cosmological constant its an integration constant. For more information on solutions in the torsion-free sector, refer to Refs. [45].

By the introduction of torsion effects with the tensors $\mathcal{B}$ and $\mathcal{A}$, the dynamical equations introduce non-trivial effects. Because of the presence of the torsion tensor, now, there are two emergent metric structures. This could be an analogue situation to the FRW universe, in principle in its universe, there is the Ricci tensor (in Einstein's Field Equations) and there is also the energy-momentum tensor, which can act as an emergent metric tensor when its no degenerate. In that sense, the Ricci tensor affine could play role of the geometrical distance (providing an affine scale factor), whereas the Poplawski metric-torsion tensor plays the role of a matter tensor, or vice-versa.

Interestingly, the system of field equations eqs. (14) to (18) present four different branches of solution, of which only one can be solved exactly. This solution, its found for the traceless part of the torsion $\psi(t)$ and it resembles the energy density from general relativity. Because of that, we can study different configurations for the α, β and γ parameters, and analyze their effects in the metric tensor coming from the Ricci. For specific values of those parameters, we are able to emulate the radiation era, the non-relativistic matter era. Additionally, we are able to define constraint for those parameters in order to secure an accelerated expansion, those constraint enable us to emulate dark matter effects.

As we see, the introduction of torsion effects in this affine model, allow us to emulate matter effects in the emergent metric structure coming from the Ricci tensor. In this formulation, the three major eras, are due to the presence of

torsion and not an energy-momentum tensor.

6 Acknowledgments

We are specially thankful to the developers of the software SageMath [52] and SageManifolds [53, 54], along with the developers of the Cadabra software [55–57], both softwares were used extensively in our calculations. Specifically, for computing the Lie derivative of the irreducible fields of the affine connection along the Killing vectors, and the variation of each term in the Polynomial Affine Gravity's action, respectively.

References

[1] A. Einstein. Die Grundlage der allgemeinen Relativitätstheorie. *Annalen der Physik*, 354(7):769–822, January 1916.

[2] A. Einstein. Die Grundlage der allgemeinen Relativitätstheorie. *Annalen der Physik*, 354(7):769–822, January 1916.

[3] Albert Einstein. Zur allgemeinen Relativitätstheorie. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, pages 778–786, January 1915.

[4] Albert Einstein. Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, pages 844–847, January 1915.

[5] Albert Einstein. Kosmologische Betrachtungen zur allgemeinen Relativitätstheorie. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, pages 142–152, January 1917.

[6] A. Friedman. Uber die krummung des raumes. *Zeitschrift für Physik*, 10(1):377–386, 1922.

[7] A. Friedmann. Uber die Moglichkeit einer Welt mit konstanter negativer Krummung des Raumes. *Zeitschrift fur Physik*, 21(1):326–332, December 1924.

[8] G. Lemaitre. A homogeneous universe of constant mass and increasing radius accounting for the radial velocity of extra-galactic nebulae. *mnras*, 91:483–490, March 1931.

[9] Georges Lemaitre. LUnivers en expansion. *Annales de la Socieacute teacute; Scientifique de Bruxelles*, 53:51, January 1933.

[10] H. P. Robertson. Kinematics and World-Structure I. *apj*, 82:284, November 1935.

[11] H. P. Robertson. Kinematics and World-Structure II. *apj*, 83:187, April 1936.

[12] H. P. Robertson. Kinematics and World-Structure III. *apj*, 83:257, May 1936.

[13] Clifford M. Will. The confrontation between general relativity and experiment. *Living Reviews in Relativity*, 17(1), June 2014.

[14] Clifford M. Will. *Theory and Experiment in Gravitational Physics*. Cambridge University Press, 2 edition, 2018.

[15] Abbott. Gw151226: Observation of gravitational waves from a 22-solar-mass binary black hole coalescence. *Physical Review Letters*, 116(24), June 2016.

[16] B. P. Abbott. Gravitational waves and gamma-rays from a binary neutron star merger: Gw170817 and grb 170817a. *The Astrophysical Journal*, 848(2):L13, October 2017.

[17] Bryce S. DeWitt. Quantum theory of gravity. i. the canonical theory. *Phys. Rev.*, 160:1113–1148, Aug 1967.

[18] Bryce S. DeWitt. Quantum theory of gravity. ii. the manifestly covariant theory. *Phys. Rev.*, 162:1195–1239, Oct 1967.

[19] S. Deser and P. van Nieuwenhuizen. One-loop divergences of quantized einstein-maxwell fields. *Phys. Rev. D*, 10:401–410, Jul 1974.

[20] S. Deser and P. van Nieuwenhuizen. Nonrenormalizability of the quantized dirac-einstein system. *Phys. Rev. D*, 10:411–420, Jul 1974.

[21] Vera C. Rubin and Jr. Ford, W. Kent. Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions. *apj*, 159:379, February 1970.

[22] Yoshiaki Sofue and Vera Rubin. Rotation curves of spiral galaxies. *Annual Review of Astronomy and Astrophysics*, 39(1):137–174, 2001.

[23] Adam G. Riess, Alexei V. Filippenko, Peter Challis, Alejandro Clocchiatti, Alan Diercks, Peter M. Garnavich, Ron L. Gilliland, Craig J. Hogan, Saurabh Jha, Robert P. Kirshner, B. Leibundgut, M. M. Phillips, David Reiss, Brian P. Schmidt, Robert A. Schommer, R. Chris Smith, J. Spyromilio, Christopher Stubbs, Nicholas B. Suntzeff, and John Tonry. Observational evidence from supernovae for an accelerating universe and a cosmological constant. *The Astronomical Journal*, 116(3):1009–1038, September 1998.

[24] H. A. Buchdahl. Non-Linear Lagrangians and Cosmological Theory. *Monthly Notices of the Royal Astronomical Society*, 150(1):1–8, 09 1970.

[25] A.A. Starobinsky. A new type of isotropic cosmological models without singularity. *Physics Letters B*, 91(1):99–102, 1980.

[26] A. A. Starobinskii. Spectrum of relict gravitational radiation and the early state of the universe. *ZhETF Pisma Redaktsiiu*, 30:719–723, December 1979.

[27] Friedrich W. Hehl, J.Dermott McCrea, Eckehard W. Mielke, and Yuval Ne'eman. Metric-affine gauge theory of gravity: field equations, noether identities, world spinors, and breaking of dilation invariance. *Physics Reports*, 258(1–2):1–171, July 1995.

[28] A. Baldazzi, O. Melichev, and R. Percacci. Metric-affine gravity as an effective field theory. *Annals of Physics*, 438:168757, March 2022.

[29] Vincenzo Vitagliano, Thomas P. Sotiriou, and Stefano Liberati. The dynamics of metric-affine gravity. *Annals of Physics*, 326(5):1259–1273, May 2011.

[30] Canan N. Karahan, Aslı Altaş, and Durmuş A. Demir. Scalars, vectors and tensors from metric-affine gravity. *General Relativity and Gravitation*, 45(2):319–343, October 2012.

[31] G. SARDANASHVILY. Classical gauge gravitation theory. *International Journal of Geometric Methods in Modern Physics*, 08(08):1869–1895, December 2011.

[32] GONZALO J. OLMO. Palatini approach to modified gravity: f(r) theories and beyond. *International Journal of Modern Physics D*, 20(04):413–462, April 2011.

[33] Arthur Stanley Eddington. *The Mathematical Theory of Relativity*. The University Press, Cambridge [Eng.], 1923.

[34] Kirill Krasnov. Pure connection action principle for general relativity. *Physical Review Letters*, 106(25), June 2011.

[35] Nikodem J. Poplawski. A unified, purely affine theory of gravitation and electromagnetism, 2007.

[36] NIKODEM J. POPŁAWSKI. On the nonsymmetric purely affine gravity. *Modern Physics Letters A*, 22(36):2701–2720, November 2007.

[37] Elie Cartan. Sur les variétés à connexion affine, et la théorie de la relativité généralisée (première partie) (Suite). *Annales scientifiques de l'École Normale Supérieure*, 3e série, 41:1–25, 1924.

[38] Elie Cartan. Sur les variétés à connexion affine, et la théorie de la relativité généralisée (deuxième partie). *Annales scientifiques de l'École Normale Supérieure*, 3e série, 42:17–88, 1925.

[39] Oskar Klein. Quantentheorie und fünfdimensionale relativitätstheorie. *Zeitschrift für Physik*, 37(12):895–906, 1926.

[40] S. Shankaranarayanan and Joseph P. Johnson. Modified theories of gravity: Why, how and what? *General Relativity and Gravitation*, 54(5), May 2022.

[41] Oscar Castillo-Felisola and Aureliano Skirzewski. A polynomial model of purely affine gravity, 2016.

[42] Oscar Castillo-Felisola and Aureliano Skirzewski. Einstein's gravity from a polynomial affine model, 2016.

[43] Oscar Castillo-Felisola, José Perdiguero, and Oscar Orellana. Cosmological solutions to polynomial affine gravity in the torsion-free sector, 2019.

[44] Oscar Castillo-Felisola. Beyond einstein: A polynomial affine model of gravity. In *Gravity - Geoscience Applications, Industrial Technology and Quantum Aspect*. InTech, feb 2018.

[45] Oscar Castillo-Felisola, José Perdiguero, Oscar Orellana, and Alfonso R Zerwekh. Emergent metric and geodesic analysis in cosmological solutions of (torsion-free) polynomial affine gravity. *Classical and Quantum Gravity*, 37(7):075013, mar 2020.

[46] Oscar Castillo-Felisola, Oscar Orellana, José Perdiguero, Francisca Ramírez, Aureliano Skirzewski, and Alfonso R. Zerwekh. Aspects of the polynomial affine model of gravity in three dimensions. *The European Physical Journal C*, 82(1), jan 2022.

[47] Oscar Castillo-Felisola, Bastian Grez, Oscar Orellana, Jose Perdiguero, Francisca Ramirez, Aureliano Skirzewski, and Alfonso R. Zerwekh. Polynomial affine model of gravity in three-dimensions. *Universe*, 8(2):68, jan 2022.

[48] Nikodem Popławski. Affine theory of gravitation. *General Relativity and Gravitation*, 46(1), dec 2013.

[49] Oscar Castillo-Felisola. Beyond einstein: A polynomial affine model of gravity. In Taher Zouaghi, editor, *Gravity*, chapter 9. IntechOpen, Rijeka, 2017.

[50] Jerzy Kijowski. On a new variational principle in general relativity and the energy of the gravitational field. *General Relativity and Gravitation*, 9(10):857–877, 1978.

[51] Oscar Castillo-Felisola, Bastian Grez, Oscar Orellana, José Perdiguero, Aureliano Skirzewski, and Alfonso R Zerwekh. Corrigendum: Emergent metric and geodesic analysis in cosmological solutions of (torsion-free) polynomial affine gravity (2020 class. quantum grav.37 075013). *Classical and Quantum Gravity*, 40(24):249501, nov 2023.

[52] The Sage Developers. *SageMath, the Sage Mathematics Software System (Version x.y.z)*, 2019. `https://www.sagemath.org`.

[53] Eric Gourgoulhon, Michal Bejger, and Marco Mancini. Tensor calculus with open-source software: the SageManifolds project. *Journal of Physics: Conference Series*, 600:012002, apr 2015.

[54] Éric Gourgoulhon and Marco Mancini. Symbolic tensor calculus on manifolds: a SageMath implementation. *Les cours du CIRM*, 6(1):1–54, 2018.

[55] Kasper Peeters. Introducing cadabra: a symbolic computer algebra system for field theory problems, 2018.

[56] Kasper Peeters. Cadabra2: computer algebra for field theory revisited. *Journal of Open Source Software*, 3(32):1118, 2018.

[57] Kasper Peeters. Cadabra: a field-theory motivated symbolic computer algebra system. *Computer Physics Communications*, 176(8):550–558, apr 2007.